

## Chapter 34 — Perinatal Urology — Numbers, Gold Standards and Traps

### IF YOU READ NOTHING ELSE

- **Urinary tract dilation (UTD):** 1–3% of pregnancies; second commonest prenatal anomaly.
- **Normal anterior–posterior diameter (APD):** <4 mm (16–27 wk), <7 mm (≥28 wk), <10 mm postnatal.
- **First postnatal US after 48 h,** once third-spaced fluid is mobilised.
- **P1 (low risk):** APD 10–<15 mm → no antibiotics, no voiding cystourethrogram (VCUG); repeat US at 3–6 mo.
- **P2 (intermediate):** prophylaxis if ureter >7 mm; otherwise at discretion.
- **P3 (high):** prophylaxis + VCUG + MAG3 (mercaptoacetyltriglycine) renogram; repeat US at 1 mo.
- **Third-trimester APD ≥15 mm** predicts UPJ (ureteropelvic junction) surgery.
- **MAG3 surgery triggers:** differential function <40%, T½ >20 min, retention on upright images.
- **Fetal LUTO intervention:** better survival (lungs), not long-term renal function.
- **Circumcision:** NNT (number needed to treat) 10 to prevent one UTI in prenatal hydronephrosis.

### PRENATAL IMAGING

- **Screening US:** 18–20 wk (ACOG, American College of Obstetricians and Gynecologists); ≈20 wk in US practice.
- **Scans per pregnancy:** low-risk ≈2; high-risk ≈4.
- **Urine production** from 14 wk; amniotic fluid = fetal urine from 16 wk.
- **Amniotic fluid index (AFI):** 4 pockets; normal 8–18; oligohydramnios <5–6.
- **Fetal MRI:** for anatomy only — cloaca, urogenital sinus, exstrophy.

### GRADING SYSTEMS

- **APD measured:** transverse mid-kidney at hilum; sagittal = classic error.
- **Prenatal APD severe:** >10 mm (2nd trimester); >15 mm (3rd).
- **Society for Fetal Urology (SFU):** grades 0–4, postnatal only; SFU 4 = parenchymal thinning.
- **SFU grades:** 1 splits sinus; 2 major calyces; 3 minor calyces.
- **SFU 3** → function preserved; **SFU 4** → early surgery.
- **UTD system:** 6 parameters; APD + SFU + ureter + bladder.
- **Pyramidal thickness ≤3 mm** → predicts pyeloplasty, reduced function.
- **Grade vs disease:** UPJ obstruction and PUV (posterior urethral valves) rise with grade; VUR (vesicoureteral reflux) does not.

### CONSULTATIONS / MYELOMENINGOCELE

- **LUTO (lower urinary tract obstruction) fetal intervention:** >40% maternal–fetal complications.
- **Fetal myelomeningocele closure** (<26 wk): ↓ VP (ventriculoperitoneal) shunt, ↑ motor; no bladder benefit.

### PRENATAL UTD AND FETAL INTERVENTION

- **A1 (low):** 4–<7 mm (16–27 wk) or 7–<10 mm (≥28 wk).
- **A2–3 (increased):** ≥7 / ≥10 mm or any abnormal parameter.
- **Follow-up:** A1 → repeat after 32 wk; A2–3 → US every 4 wk.
- **Oligohydramnios <27 wk** → poor outcome (pulmonary hypoplasia).
- **Poor prognosis:** cortical cysts, echogenic kidneys, unfavourable urine.
- **Intervention window:** 18–24 wk; need ≈1 in 60,000.
- **Fetal urine markers:** Na, Cl, osmolality, β2-microglobulin — unvalidated.
- **ERKNet 2022:** fluid, echogenicity, urine biomarkers don't predict renal function.
- **Ruano stage II** (preserved function) = the stage to intervene.
- **PLUTO (Percutaneous shunting in LUTO) trial:** shunt ↑ survival; no renal benefit; 44% complications; stopped early.

- **Fetal cystoscopy:** diagnoses + treats; only it improved renal function in PUV.
- **Cystoscopy fistula** (urethrocuteaneous/rectourethral): up to 10%.
- **Rodeck shunt:** lies flush → less migration than Harrison.
- **Bilateral renal agenesis:** serial amnioinfusion (RAFT trial).

### POSTNATAL EVALUATION

- **P2/P3 APD:** ≥15 mm; P2 = peripheral calyces/ureter; P3 = parenchyma/bladder.
- **Resolution:** SFU 1 → 100% by ≈2.5 yr; SFU 2 → 95% by 4 yr.
- **VUR:** up to 31% of prenatal UTD; 25% have normal first US.
- **VCUG as inpatient** if bladder wall ≥0.5 cm, distended bladder, bilateral hydroureteronephrosis.
- **VCUG technique** (AAP): non-balloon catheter, bladder cycling, ALARA.
- **MAG3 vs DTPA:** MAG3 is gold standard; ≈50% tubular extraction per pass.
- **DMSA (dimercaptosuccinic acid):** cortical defects, MCDK (multicystic dysplastic kidney) non-function, duplex polar function.
- **CeVUS (contrast voiding urosonography):** no radiation, sensitive for VUR; poor for urethra.

### MANAGEMENT BY RISK / UTI

- **UTI incidence:** 8–22%; ≈90% occur in first year.
- **UTI risk factors:** female, uncircumcised, high-grade dilation (± ureter, VUR).
- **P3 prophylaxis duration:** until UPJ surgery or 1 year.
- **Antibiotics from birth:** amoxicillin, first-generation cephalosporin.
- **Nitrofurantoin/trimethoprim:** only from ≥8 wk.
- **Bilateral dilation + oligohydramnios** → decompress, prophylaxis, evaluate.

### KEY CONDITIONS

- **Commonest cause:** transient UTD 50–70%; UPJ 10–30%; VUR 10–40%.
- **LUTO share:** 0.1–0.5% of prenatal UTD.
- **Commonest surgical cause:** UPJ obstruction.
- **APD 15 mm:** sensitivity ≈0.8, specificity ≈0.8; postnatal APD predicts better.
- **Likely DRF >40%:** P2, APD <20 mm, pyramidal thickness >3 mm.
- **Observed UPJ losing function:** APD 20–40 mm (Great Ormond Street trial).
- **Operate UPJ if:** rising APD, DRF <40% + delayed drainage, falling DRF, stone/infection.

- **Prenatal VUR:** male, bilateral, high grade, dysplasia (up to 40%).
- **Screen VUR:** girls with high-grade UTD, uncircumcised boys; not circumcised low-grade.
- **Megaureter surgery risk:** ureter  $\geq 13$  mm, DRF  $< 30\%$ .
- **BAPU megaureter:** prenatal ureter  $> 7$  mm after 30 wk  $\rightarrow$  postnatal RUS, prophylaxis, VCUG.
- **BAPU surgery:** DRF  $< 40\%$ , febrile UTI, pain.
- **Megaureter follow-up:** into adulthood (late silent deterioration).
- **Megaureter under 1 yr:** temporise; definitive reimplant after 1 yr.

## LUTO

- **PUV:** 1.4–2.2 per 10,000; commonest LUTO.
- **PUV neonatal drainage:** 5 Fr feeding tube; confirm on KUB (kidney–ureter–bladder) film.
- **Avoid balloon catheter** (posterior urethral obstruction).
- **Prune-belly syndrome (PBS):** absent abdominal muscles + impalpable testes + megacystis.
- **Keyhole sign + pop-offs** (ascites, urinoma)  $\rightarrow$  PUV/urethral atresia.
- **Urethral atresia:** catheter won't pass  $\rightarrow$  vesicostomy  $\rightarrow$  PADUA.
- **No oligohydramnios**  $\rightarrow$  megacystis, bilateral VUR, MMIHS.
- **MMIHS** (megacystis–microcolon): non-obstructed; prophylaxis only if recurrent UTI.
- **Post-decompression anuria** (thick bladder)  $\rightarrow$  antimuscarinics.
- **Renal insufficiency**  $\rightarrow$  no immediate VCUG.

## KIDNEY ANOMALIES

- **MCDK:** VUR  $\approx 20\%$ ; no Wilms risk; no prophylaxis.
- **Solitary kidney:** check BP, proteinuria, yearly US to puberty.
- **Remove MCDK** only if persistent with solid component.
- **Meyer–Weigert:** upper pole  $\rightarrow$  obstructs/ectopic; lower pole  $\rightarrow$  refluxes.
- **Suspect duplex:** ureter more dilated than pelvis.
- **Ureterocele + high-grade VUR**  $\rightarrow$  ureterocele excision in 96%.
- **Ureterocele on US:** thin intravesical cyst; ectopic ureter shows two walls.
- **No VUR**  $\rightarrow$  upper-pole surgery alone suffices.

## EXSTROPHY / CLOACA / CAH

- **Bladder exstrophy:** detected prenatally  $\approx 50\%$ ; not an emergency.
- **Cloacal exstrophy (OEIS):** major criteria + omphalocele + lumbosacral anomaly.
- **Persistent cloaca:** 1 in 50,000; females; calcified meconium.
- **Congenital adrenal hyperplasia (CAH):** 21-hydroxylase gene on chromosome 6p.
- **CAH frequency:** 1 in 15,000–16,000; salt-wasting 75%.
- **CAH carrier pregnancies:** 1 in 8 affected with ambiguity.
- **Cloaca + polyhydramnios**  $\rightarrow$  suspect tracheo-oesophageal fistula.

- **Normal genitalia + trapped vaginal urine**  $\rightarrow$  urogenital sinus, not CAH.

## PERINATAL EMERGENCIES

- **Urinary ascites:** up to 30% of neonatal ascites; commonest cause PUV.
- **Seldinger drainage:** 16G  $\rightarrow$  3.5 Fr; 14G  $\rightarrow$  5 Fr feeding tube.
- **Solid tumour  $< 6$  mo:** mesoblastic nephroma ( $\uparrow$  BP, polyhydramnios).
- **Adrenal haemorrhage:** right side; 10–15% bilateral; no Doppler flow; eggshell calcification.
- **Ovarian cysts:** up to 30% of girls; resolve by 6 mo.
- **Hydrometrocolpos:** can mimic distended bladder.
- **Urethral prolapse:** midline collar; Black girls; conservative care.
- **Anorectal malformation:** genitourinary anomalies up to 75%.
- **Anorectal malformation work-up:** spinal US; videourodynamics at 6 wk; prophylaxis.
- **Commonest cause of ambiguous genitalia:** CAH (46,XX DSD).
- **Ambiguous genitalia:** check electrolytes first; defer gender assignment.
- **First void:** within 24 h; after circumcision within 8 h.
- **Renal vein thrombosis (RVT):** up to 20% of gross haematuria.
- **RVT triad:** haematuria + flank mass + thrombocytopenia.
- **RVT prothrombotic disorder:** up to 50%.
- **Renal artery thrombosis:** umbilical artery catheter  $\rightarrow$  hypertension; remove catheter.
- **Cortical necrosis:** lost corticomedullary differentiation; lifelong renal-failure risk.
- **Torsion salvage:** postnatal  $\leq 44\%$ ; prenatal  $\approx 0$ .
- **Prenatal torsion**  $\rightarrow$  urgent contralateral orchidopexy.

## CLASSIC TRAPS

- **Trap:** APD 10 mm always severe — Actually: severe in 2nd trimester, moderate in 3rd.
- **Trap:** dilation grade predicts VUR — Actually: no trend; VUR at any grade.
- **Trap:** US at 24 h — Actually: wait  $> 48$  h; earlier scan under-reads dilation.
- **Trap:** APD  $\geq 15$  mm = P3 — Actually: parenchymal/bladder abnormality defines P3.
- **Trap:** fetal shunt saves kidneys — Actually: saves lungs/survival; renal outcome unchanged.
- **Trap:** MCDK needs nephrectomy for Wilms risk — Actually: no Wilms link; observe.
- **Trap:** balloon catheter for PUV — Actually: 5 Fr feeding tube, KUB-confirmed.
- **Trap:** fetal myelomeningocele closure improves bladder — Actually: urodynamics and CIC need unchanged.
- **Trap:** nitrofurantoin/trimethoprim in newborn — Actually: amoxicillin or cephalosporin until 8 wk.
- **Trap:** bloody diaper = haematuria — Actually: confirm on urinalysis; oestrogen withdrawal, urates.